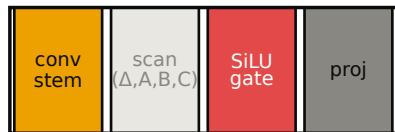


Step 1 -- Freeze the scan (partial and full)

conv stem scan (Δ, A, B, C) SiLU gate proj

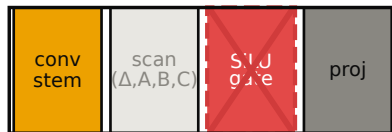
BM3-frozen (partial) / frozen-all (full)



partial / full freeze vs S4D:
+12-18pp / +20-23pp (forward)
→ input-indep. coefficients suffice

Step 2 -- Remove the gate (in the frozen host)

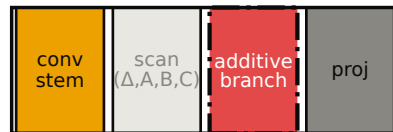
frozen – gate



partial host, forward: +9/+16/+27pp
(resolved at –6 dB only)
reverse: resolved at every level

Step 3 -- Replace the gate by an additive branch (in the host)

*frozen + additive branch
replaced*



mult. – add. at –6 dB:
+23 (full) / +27 (partial) pp
additive vs gateless: not resolved

Conditional on this block, testbed and budget: input-independent scan coefficients suffice; gate cost grows with noise only in the forward fully frozen block.